

Voluntary Report – Voluntary - Public Distribution

Date: October 06,2020

Report Number: IN2020-0140

Report Name: Record-setting Southwest Monsoon Season Ends

Country: India

Post: Mumbai

Report Category: Agricultural Situation, Agriculture in the News, Agriculture in the Economy, Climate Change/Global Warming/Food Security, Coffee, Cotton and Products, Grain and Feed, Oilseeds and Products, Sugar

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Report Highlights:

The southwest monsoon has receded in most parts of the country, ending the season (Jun/Sep) with cumulative rainfall nine percent higher than the normal fifty-year average of 881 millimeters. This year experienced the third highest seasonal rainfall level since 1990, after 1994 and last year. Planting area for the kharif crop is one percent higher than last year, with farmers dedicating significant area to crops that have assured central/state government procurement.

Southwest Monsoon Update

According to the Indian Meteorological Department (IMD), the cumulative rainfall for the southwest monsoon 2020 (Jun/Sep) is nine percent higher than normal fifty-year average of 881 millimeters. Since 1990, this was the third highest year of seasonal rainfall after 1994 and last year. The southwest monsoon in 2019 and 2020 are the two consecutive years of above normal monsoon rainfall since 1958 and 1959.

The rainfall distribution during the monsoon varied widely on a monthly basis. The rainfall over the country was 118 percent, 90 percent, 127 percent, and 104 percent of the long period average (LPA) during June, July, August and September, respectively. In June, the timely onset and twelve-day advance coverage (against the normal date) throughout the entire country of good monsoon rains were followed by a weakening of the monsoon, which lead to deficit rains in July. In August, the monsoon revived and caused 2-3 instances of riverine floods in Odisha, Telangana, Madhya Pradesh, south Gujarat, and south Rajasthan. August 2020 experienced record rainfall, with nationwide rain levels at 127 percent of the LPA. It was highest rainfall in the last 44 years since August 1976. During August 2020, consecutively for 4 weeks, India received excess rain around 13 to 41 percent above LPA between August 12 – September 2, 2020. Excess rains in August were followed by deficit rains in Northwest India, which were offset by excess rains in Central India. For more detail please refer to [IMD press release - Salient features of Monsoon 2020](#). Overall, more than 75 percent of the districts across India have received normal to excess rains during the monsoon.

Delayed withdrawal of Monsoon, and onset of Northeast Monsoon

According to IMD, the southwest monsoon withdrew from northwest India on September 28, 2020 compared to the normal date of September 17, 2020. This withdrawal has been delayed primarily due to active monsoon trough along with the formation of two low pressure systems in September 2020. As of October 1, the southwest monsoon has withdrawn from Punjab, western Himalayan region, Haryana, Chandigarh, Delhi, many parts for Rajasthan, and some parts of Uttar Pradesh. As the monsoon withdraws, widespread rains are forecast over West Bengal, Odisha, Bihar, Jharkhand, and Andhra Pradesh during the next five days (Oct 1-5). According to IMD, below-normal rainfall is likely during the Northeast Monsoon (October/ December) over the southern parts of the South Asia including some parts of southeastern India.

First 2020/21 Advance Area Estimates for Kharif 2020/21 published

On October 1, the Ministry of Agriculture and Farmers Welfare's (MOAFW) published the first advance area estimates for Kharif 2020/21 season and the fourth advance area estimates for Kharif 2019/20 season. The 2020/21 estimates forecast a one percent increase in overall planted area from last year, and a two percent increase over the five-year average. The estimates also forecast a three percent increase in area for rice and oilseeds, nine percent in area under sugarcane (albeit from a smaller base), and one percent increase in area under pulses, while coarse cereals and cotton are expected to be decline by four percent and three percent, respectively. Farmers have preferred to increase area for crops (such as rice, grains and groundnut) that have assured central/state government procurement under the minimum support price (MSP) program. For more details, please refer to [All India Crop Situation dated October 1, 2020](#).

Reservoir Storage

The Central Water Commission monitors the live storage status of 123 reservoirs around the country on a weekly basis. As per the reservoir storage bulletin dated October 1, 2020, the live storage available in these reservoirs is 148.247 billion cubic meters (BCM), which is 87 percent of total live storage capacity. The live storage available in these reservoirs for the corresponding period last year was 151.071 BCM (88 percent), and the average of the last 10 years live storage was 129.953 BCM (76 percent). As such, the current storage position is less than the corresponding period last year, but better than the average storage level of the last ten years during the corresponding period.

Out of 123 reservoirs, 119 reservoirs reported more than 80 percent of normal storage levels and 4 reservoirs reported 80 percent or below of normal storage. Out of these 4 reservoirs, 1 has 51-60 percent of normal storage. According to the Central Water Commission, normal storage represents the average storage level of last ten years, close to normal storage represents a shortfall of up to 20 percent of normal, deficient storage is where the shortfall is greater than 20 percent of the normal and up to 60 percent of the normal, highly deficient means shortfall is more than 60 percent of normal.

States that have better storage (in percentage) than last year for the corresponding period include Jharkhand, West Bengal, Tripura, Maharashtra, Uttar Pradesh, Uttarakhand, Madhya Pradesh, Chhattisgarh, Kerala and Telangana. For more details please refer to the [Reservoir Storage Bulletin as on October 1, 2020](#).

**Table 1. India: Southwest Monsoon Regional Rainfall Distribution
from June 1 to September 30, 2020**

Regions	2020 Actual (mm)	Normal (mm)*	2020 Percentage Departure from Normal
Northwest India	505.7	599.5	-16%
Central India	1,123.8	976.6	15%
Southern Peninsula	939.9	726.2	29%
East and Northeast India	1,500.3	1,410.4	6%
All India	957.6	880.6	9%

*Normal Rainfall is the fifty-year average from 1951-2000

Source: Indian Meteorological Department

Table 2. India: *Kharif* 2020 Sown Area (in million hectares)

Crop	First Advance Estimates 2020/21	Fourth Advance Estimates 2019/20	Normal Area for Kharif Season	Y-o-Y Change	Change from Normal
Rice	40.105	39.019	39.729	3%	1%
Pulses	13.515	13.404	12.888	1%	5%
Coarse Cereals	17.617	18.378	18.485	-4%	-5%
Oilseeds	19.661	19.154	18.139	3%	8%
Sugarcane	4.993	4.567	4.846	9%	3%
Jute and Mesta	0.69	0.675	0.76	2%	-9%
Cotton	12.957	13.369	12.227	-3%	6%
Total	109.538	108.566	107.074	1%	2%

Source: Ministry of Agriculture and Farmers Welfare, Government of India

**Normal Area is the five-year average of the area from 2014-2018

RAINFALL STATISTICS - MONSOON 2020

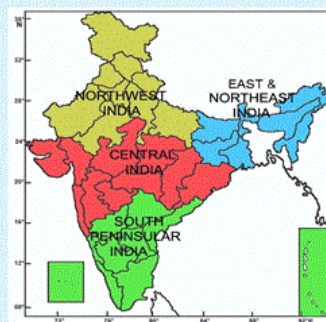
JUNE - 2020	1-Jun	TO	30-Jun
REGION	ACTUAL	NORMAL	% DEP
COUNTRY AS A WHOLE	196.2	166.9	17.6
NORTHWEST INDIA	77.9	75.3	3.5
EAST & NORTHEAST INDIA	401.4	347.1	15.7
CENTRAL INDIA	220.9	169.2	30.5
SOUTH PENINSULA	172.6	160.2	7.7

JULY - 2020	1-Jul	TO	31-Jul
REGION	ACTUAL	NORMAL	% DEP
COUNTRY AS A WHOLE	257.6	285.3	-9.7
NORTHWEST INDIA	157.7	212.2	-25.7
EAST & NORTHEAST INDIA	471.8	432.0	9.2
CENTRAL INDIA	250.7	322.8	-22.3
SOUTH PENINSULA	251.7	216.7	16.2

AUG - 2020	1-Aug	TO	31-Aug
REGION	ACTUAL	NORMAL	% DEP
COUNTRY AS A WHOLE	327.0	258.2	26.6
NORTHWEST INDIA	212.3	202.7	4.8
EAST & NORTHEAST INDIA	279.9	346.0	-19.1
CENTRAL INDIA	495.2	307.3	61.2
SOUTH PENINSULA	256.2	189.2	35.4

SEP - 2020	1-Sep	TO	30-Sep
REGION	ACTUAL	NORMAL	% DEP
COUNTRY AS A WHOLE	177.3	170.2	4.2
NORTHWEST INDIA	58.9	109.2	-46.0
EAST & NORTHEAST INDIA	347.2	285.2	21.7
CENTRAL INDIA	156.9	177.2	-11.5
SOUTH PENINSULA	259.8	160.0	62.4

CUMULATIVE SEASONAL RAINFALL	1-Jun	TO	30-Sep
REGION	ACTUAL	NORMAL	% DEP
COUNTRY AS A WHOLE	957.6	880.6	8.7
NORTHWEST INDIA	505.7	599.5	-15.6
EAST & NORTHEAST INDIA	1500.3	1410.4	6.4
CENTRAL INDIA	1123.8	976.6	15.1
SOUTH PENINSULA	939.9	726.2	29.4



FOUR HOMOGENEOUS REGIONS OF INDIA

(Based on real time data)

Fig 1. All India seasonal Monsoon rainfall in percentage departure from normal

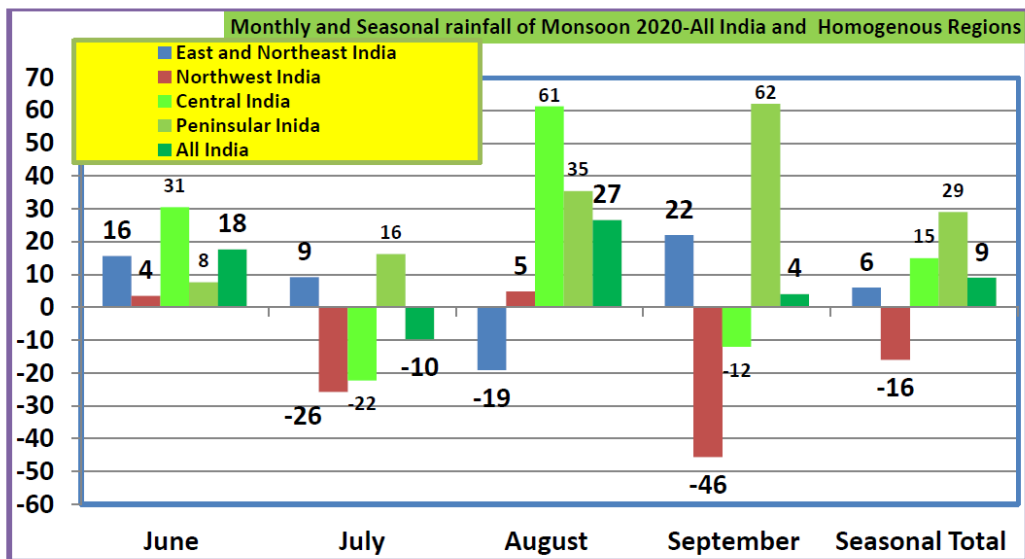
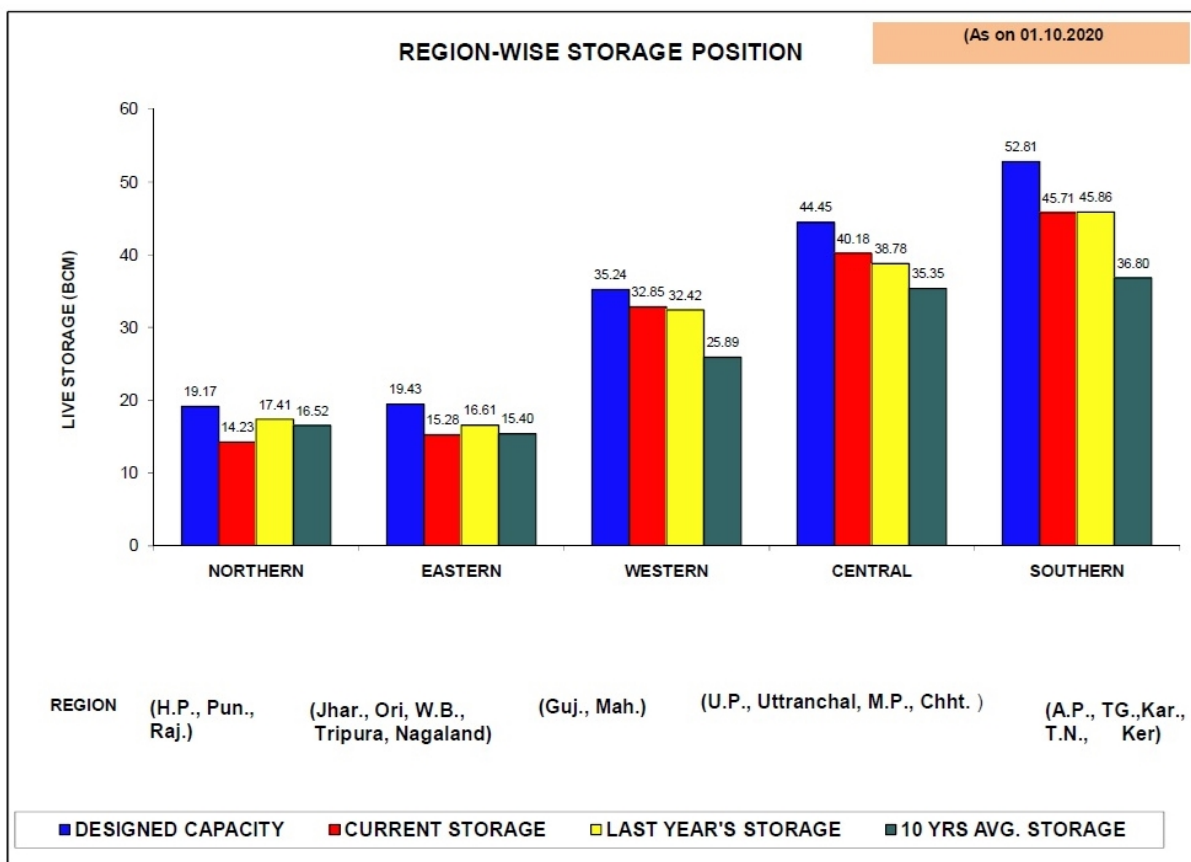


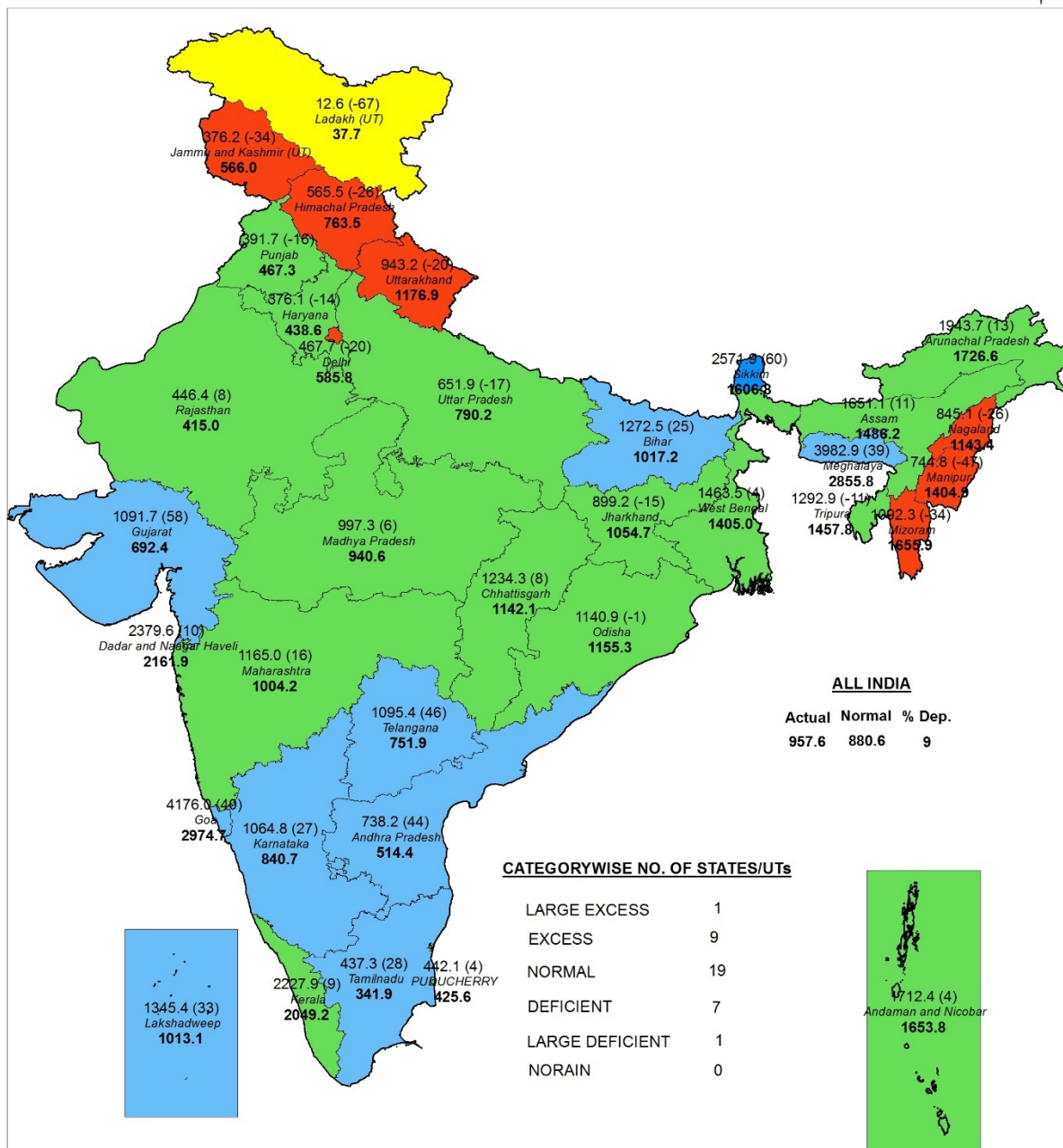
Fig 2. Monthly and Seasonal monsoon rainfall of 2020 over Broad homogenous region and Country as a whole in percentage departure from normal





STATE RAINFALL MAP

Period : 01-06-2020 To 30-09-2020



Legend

Large Excess [60% or more] Excess [20% to 59%] Normal [-19% to 19%] Deficient [-59% to -20%] Large Deficient [-99% to -60%] No Rain [-100%] No Data

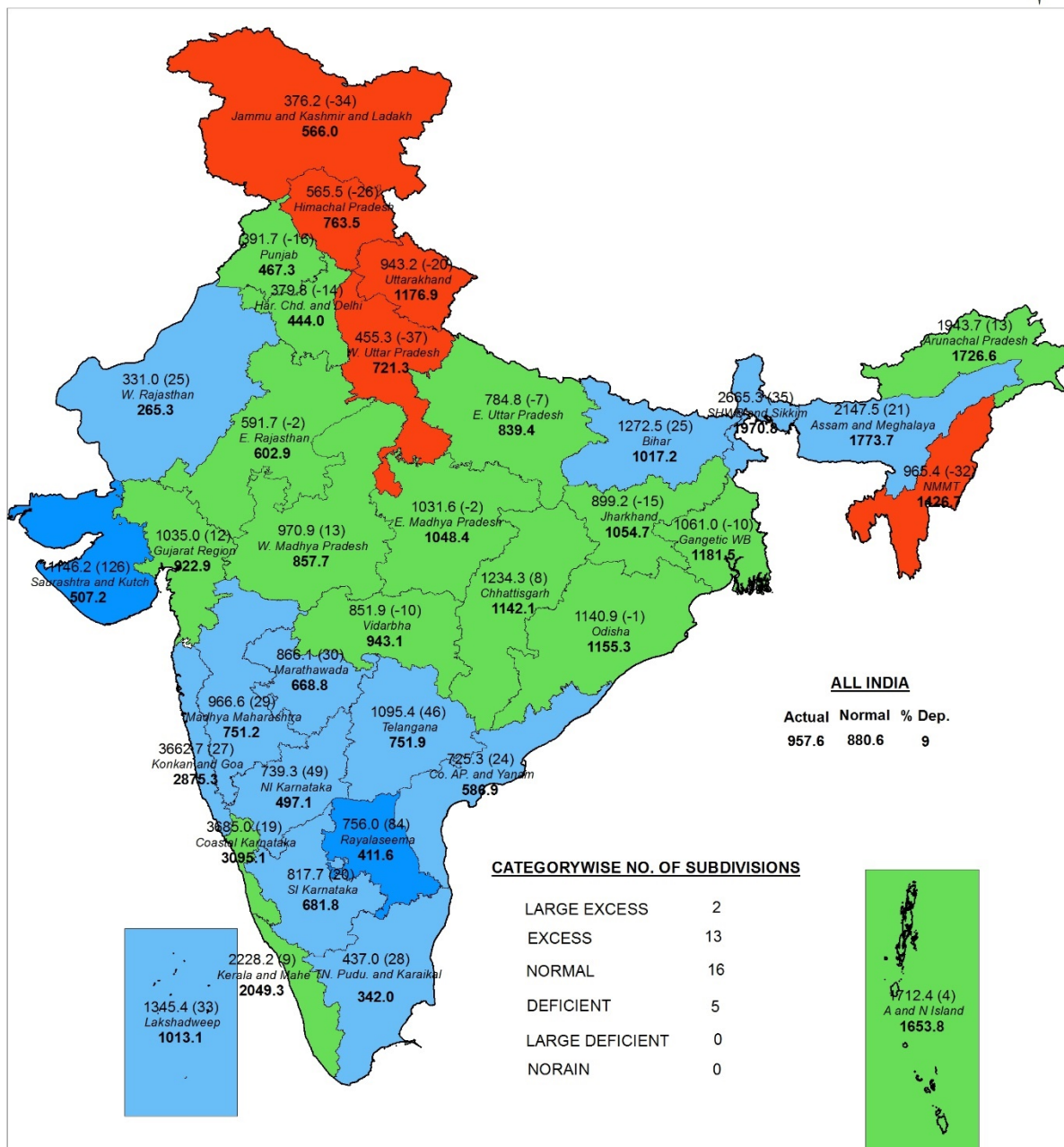
NOTES :

- RainFall figures are based on operation data.
- Small figures indicate actual rainfall (mm), while bold figures indicate Normal rainfall (mm).
- Percentage Departures of rainfall are shown in brackets.



SUBDIVISION RAINFALL MAP

Period : 01-06-2020 To 30-09-2020



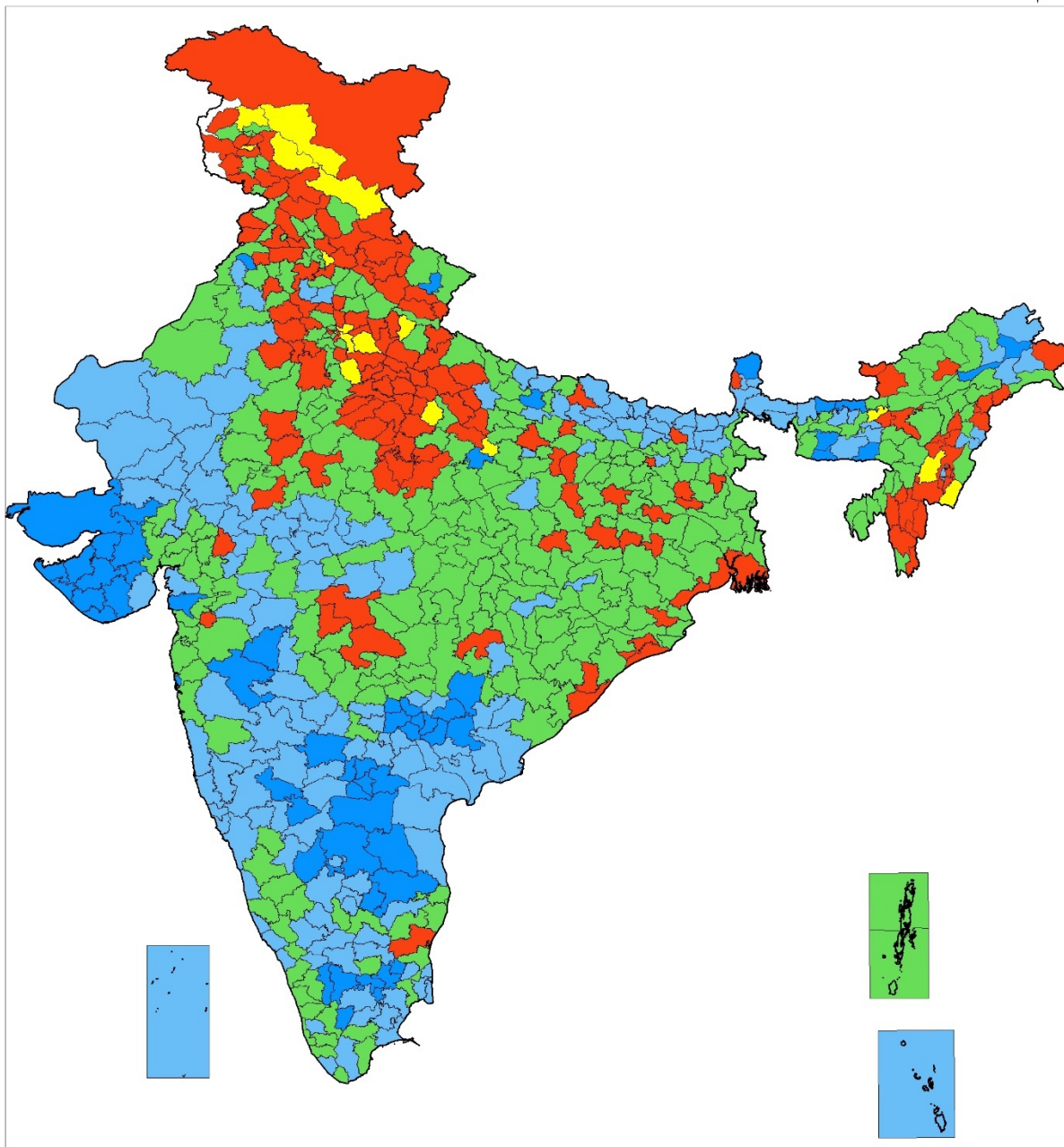


भारत मौसम विज्ञान विभाग
INDIA METEOROLOGICAL DEPARTMENT

जल मौसम विज्ञान प्रभाग, नई दिल्ली
HYDROMET DIVISION, NEW DELHI

DISTRICT RAINFALL MAP

Period : 01-06-2020 To 30-09-2020



Legend

Large Excess [60% or more] Excess [20% to 59%] Normal [-19% to 19%] Deficient [-59% to -20%] Large Deficient [-99% to -60%] No Rain [-100%] No Data

NOTES :

a) RainFall figures are based on operation data.



India Meteorological Department
Hydromet Division, New Delhi

STATE-WISE RAINFALL DISTRIBUTION

		Week:24-09-2020 To 30-09-2020				Period:01-06-2020 To 30-09-2020			
S NO	MET. SUBDIVISION/UT/STATE/DISTRICT	ACTUAL (mm)	NORMAL (mm)	%DEP.	CAT.	ACTUAL (mm)	NORMAL (mm)	% DEP.	CAT.
REGION : EAST AND NORTH EAST INDIA									
1	ARUNACHAL PRADESH	161.6	72.2	124%	LE	1943.7	1726.6	13%	N
2	ASSAM	87.3	55.6	57%	E	1651.1	1486.2	11%	N
3	MEGHALAYA	595.3	93.1	539%	LE	3982.9	2855.8	39%	E
4	NAGALAND	30.8	46.8	-34%	D	845.1	1143.4	-26%	D
5	MANIPUR	23.8	61.8	-62%	LD	744.8	1404.9	-47%	D
6	MIZORAM	39.3	79.2	-50%	D	1092.3	1655.9	-34%	D
7	TRIPURA	71.0	54.1	31%	E	1292.9	1457.8	-11%	N
8	SIKKIM	51.3	63.2	-19%	N	2571.9	1606.8	60%	LE
9	WEST BENGAL	69.2	75.0	-8%	N	1463.5	1405.0	4%	N
10	JHARKHAND	23.6	47.5	-50%	D	899.2	1054.7	-15%	N
11	BIHAR	133.0	46.9	184%	LE	1272.5	1017.2	25%	E
REGION : NORTH WEST INDIA									
1	UTTAR PRADESH	46.1	20.0	131%	LE	651.9	790.2	-17%	N
2	UTTARAKHAND	11.2	22.5	-50%	D	943.2	1176.9	-20%	D
3	HARYANA	2.1	7.2	-71%	LD	376.1	438.6	-14%	N
4	CHANDIGARH (UT)	1.6	23.0	-93%	LD	791.1	846.5	-7%	N
5	DELHI (UT)	0.0	11.7	-100%	NR	467.7	585.8	-20%	D
6	PUNJAB	0.3	14.2	-98%	LD	391.7	467.3	-16%	N
7	HIMACHAL PRADESH	1.4	18.0	-92%	LD	565.5	763.5	-26%	D
8	JAMMU & KASHMIR (UT)	1.1	19.2	-94%	LD	376.2	566.0	-34%	D
9	LADAKH (UT)	0.0	1.0	-100%	NR	12.6	37.7	-67%	LD
10	RAJASTHAN	7.2	6.6	10%	N	446.4	415.0	8%	N
REGION : CENTRAL INDIA									
1	ODISHA	19.3	36.8	-47%	D	1140.9	1155.3	-1%	N
2	MADHYA PRADESH	28.1	19.1	47%	E	997.3	940.6	6%	N
3	GUJARAT	11.4	16.3	-30%	D	1091.7	692.4	58%	E
4	DADAR & NAGAR HAVELI (UT)	41.0	86.4	-53%	D	2379.6	2161.9	10%	N
5	DAMAN & DIU (UT)	29.1	62.1	-53%	D	1686.2	1611.2	5%	N
6	GOA	128.4	64.6	99%	LE	4176.0	2974.7	40%	E
7	MAHARASHTRA	24.0	37.2	-36%	D	1165.0	1004.2	16%	N
8	CHHATTISGARH	13.6	29.3	-54%	D	1234.3	1142.1	8%	N
REGION : SOUTH PENINSULA									
1	ANDAMAN & NICOBAR (UT)	131.2	111.7	17%	N	1712.4	1653.8	4%	N
2	ANDHRA PRADESH	62.7	38.5	63%	LE	738.2	514.4	44%	E
3	TELANGANA	62.5	32.4	93%	LE	1095.4	751.9	46%	E
4	TAMIL NADU	36.7	32.4	13%	N	437.3	341.9	28%	E
5	PUDUCHERRY (UT)	67.5	36.3	86%	LE	442.1	425.6	4%	N
6	KARNATAKA	40.9	42.5	-4%	N	1064.8	840.7	27%	E
7	KERALA	21.2	67.5	-69%	LD	2227.9	2049.2	9%	N
8	LAKSHADWEEP (UT)	6.2	36.1	-83%	LD	1345.4	1013.1	33%	E
COUNTRY :		38.6	31.4	23%		957.6	880.6	9%	

CATEGORYWISE DISTRIBUTION OF NO.OF STATES

CATEGORY	Week:24-09-2020 To 30-09-2020	Period:01-06-2020 To 30-09-2020
	NO.OF STATES	NO.OF STATES
Large Excess	8	1
Excess	3	9
Normal	6	19
Deficient	10	7
Large Deficient	8	1
NoRain	2	0
NoData	0	0

PROGRESS OF MONSOON 2020 WEEK BY WEEK

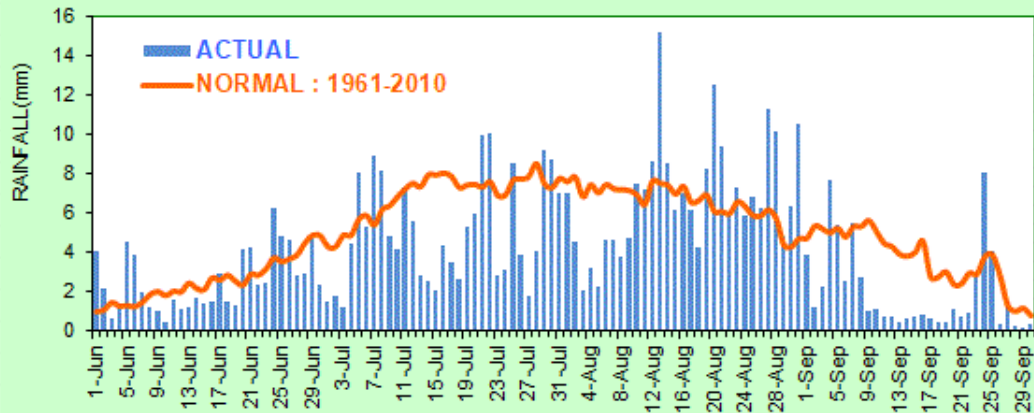
S.NO	MET.SUBDIVISION	WEEK ENDINGS																	
		3-Jun	10-Jun	17-Jun	24-Jun	1-Jul	8-Jul	15-Jul	22-Jul	29-Jul	5-Aug	12-Aug	19-Aug	26-Aug	2-Sep	9-Sep	16-Sep	23-Sep	30-Sep
1	A & N ISLANDS																		
2	ARUNACHAL PRADESH																		
3	ASSAM & MEGHALAYA																		
4	NAG., MANI., MIZO. & TRIPURA																		
5	S.H.W.B. & SIKKIM																		
6	GANGATIC W.B.																		
7	ODISHA																		
8	JHARKHAND																		
9	BIHAR																		
10	EAST U.P.																		
11	WEST U.P.																		
12	UTTARAKHAND																		
13	HAR., CHANDI. & DELHI																		
14	PUNJAB																		
15	HIMACHAL PRADESH																		
16	J & K AND LADAKH																		
17	WEST RAJASTHAN																		
18	EAST RAJASTHAN																		
19	WEST M.P.																		
20	EAST M.P.																		
21	GUJARAT REGION																		
22	SAURASHTRA & KUTCH																		
23	KONKAN & GOA																		
24	MADHYA M'RASHTRA																		
25	MARATHAWADA																		
26	VIDARBHA																		
27	CHATTISGARH																		
28	COASTAL A. P. & YANAM																		
29	TELANGANA																		
30	RAYALASEEMA																		
31	TAMIL., PUDU. & KARAIKAL																		
32	COASTAL KARNATAKA																		
33	N.I.KARNATAKA																		
34	S.I.KARNATAKA																		
35	KERALA & MAHE																		
36	LAKSHADWEEP																		

LEGEND:

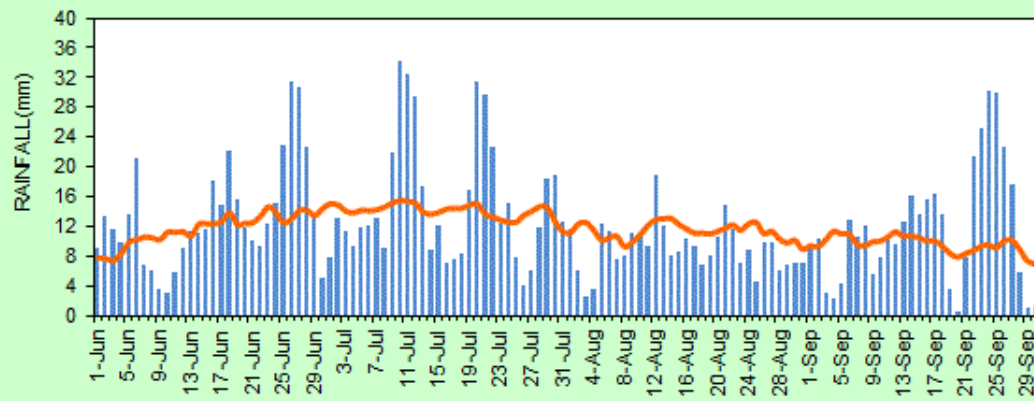
 LARGE EXCESS +60% OR MORE	 EXCESS +20% TO +59%	 NORMAL +19% TO -19%
 DEFICIENT -20% TO -59%	 LARGE DEFICIENT -60% OR LESS	 NO RAIN

(वास्तविक समय के आंकड़ों पर आधारित)

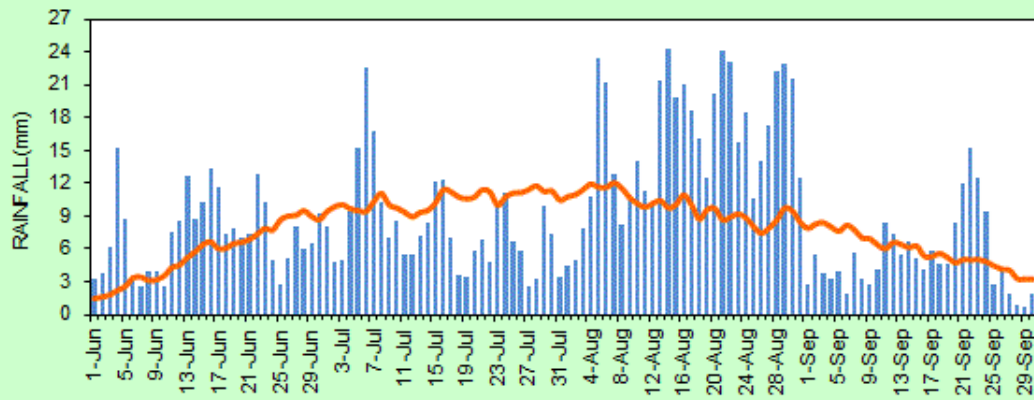
(2020)DAILY MEAN RAINFALL(mm) OVER THE FOUR HOMOGENEOUS REGIONS



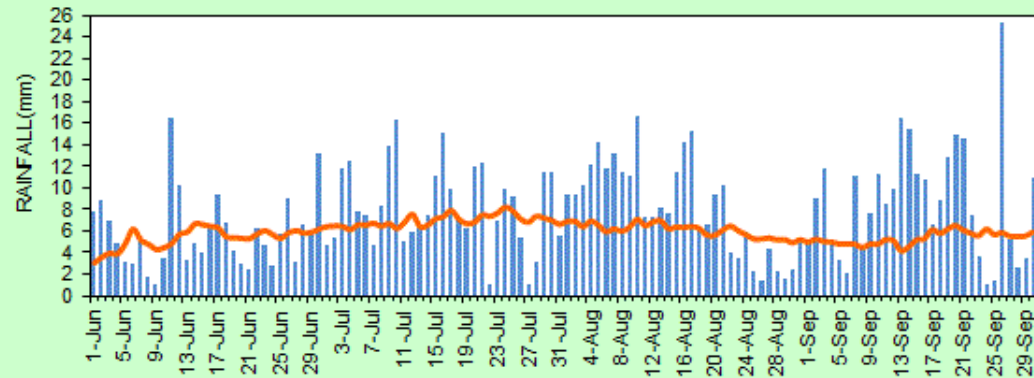
NORTH WEST INDIA



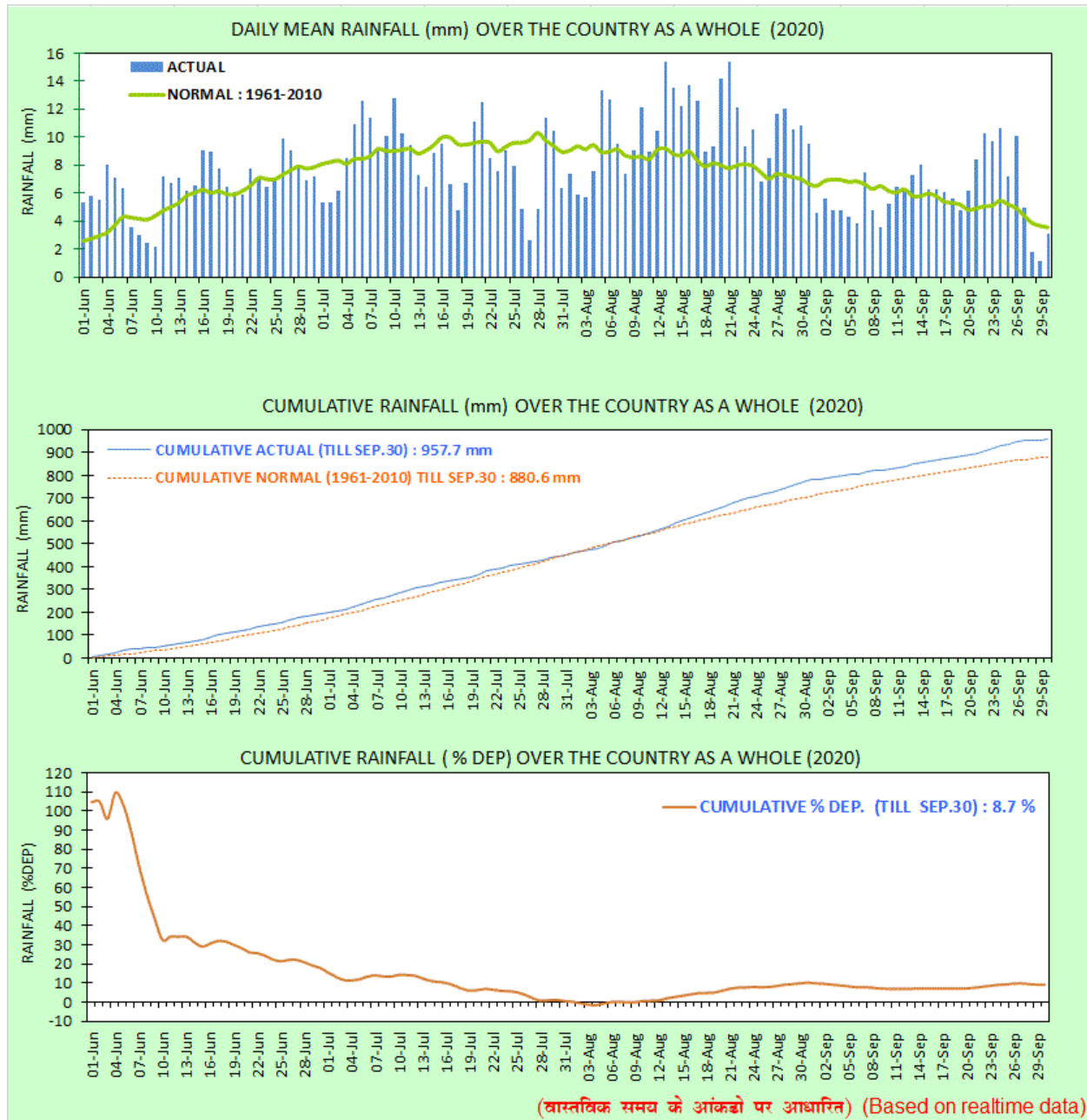
EAST & NORTHEAST INDIA



CENTRAL INDIA



SOUTH PENINSULAR INDIA



Attachments:

No Attachments.